VI. Calculation based on % composition

Calculate % of S in H_2SO_4

molecular mass of H_2SO_4

$= (1 \times 2) + (32 \times 1) + (16 \times 4)$

$= 2 + 32 + 64$

$= 98$ g

$$\% \text{ of S in } H_2SO_4 = \frac{\text{Mass of sulphur}}{\text{Molecular mass of } H_2SO_4} \times 100$$

$$\% \text{ of S in } H_2SO_4 = \frac{32}{98} \times 100$$

$= 32.65\ \%$

Points to Remember

- Two or more forms of an element having the same atomic number, but different mass number are called Isotopes ($_{17}Cl^{35}$, $_{17}Cl^{37}$).
- Atoms of different elements having the same mass number, but different atomic numbers are called Isobars ($_{18}Ar^{40}$, $_{20}Ca^{40}$).
- Atoms of different elements having the same number of neutrons, but different atomic number and different mass number are called Isotones ($_{6}C^{13}$, $_{7}N^{14}$).
- Relative atomic mass of an element is the ratio between the mass of one atom of the element to 1/12th of the mass of the atom of carbon -12.
- Average atomic mass of an element is calculated by adding the masses of its isotopes, each multiplied by their natural abundance on the Earth.
- Relative molecular mass of a molecule is the ratio between the mass of one molecule of the substance to 1/12th of the mass of the atom of carbon – 12.
- The Avogadro's law states that "equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules".
- The vapour density is defined as "the ratio between the masses of equal volumes of a gas (or a vapour) and hydrogen under the same condition".
- Atomicity of a monoatomic element = Molecular mass / Atomic Mass.
- Molecular mass = 2 × Vapour density.

TEXTBOOK EVALUATION

T97QU

I. Choose the best answer.

1. Which of the following has the smallest mass?
   a. 6.023×10^{23} atoms of He
   b. 1 atom of He
   c. 2 g of He
   d. 1 mole atoms of He

2. Which of the following is a triatomic molecule?
   a. Glucose b. Helium
   c. Carbon dioxide d. Hydrogen

3. The volume occupied by 4.4 g of CO_2 at S.T.P
   a. 22.4 litre b. 2.24 litre
   c. 0.24 litre d. 0.1 litre

4. Mass of 1 mole of Nitrogen atom is
   a. 28 amu b. 14 amu
   c. 28 g d. 14 g

5. Which of the following represents 1 amu?
   a. Mass of a C – 12 atom
   b. Mass of a hydrogen atom
   c. $1/12^{th}$ of the mass of a C – 12 atom
   d. Mass of O – 16 atom

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